

E-Commerce Sales Data Analysis — Project Report

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Course: Data Analytics

1. Introduction

E-commerce businesses generate large amounts of sales data. Analysing this data helps businesses understand customers, products, regions and sales trends. This project demonstrates a complete beginner-friendly data analytics workflow using Python.

2. Problem Statement

The objective is to analyse e-commerce transaction data and extract useful business insights from sales, profit, category, product, region and time-based information.

3. Objectives

1. Understand and inspect the dataset.
2. Check data quality and missing values.
3. Calculate sales and profit KPIs.
4. Compare category and regional performance.
5. Identify high-performing products.
6. Analyse monthly sales trends.
7. Create visualizations to communicate findings.

4. Dataset

The project uses a synthetic dataset containing 500 e-commerce orders from 2025. Important fields include Order ID, Order Date, Customer ID, Region, Category, Product, Quantity, Unit Price, Discount, Sales, Cost, Profit and Payment Method.

5. Methodology

The workflow is:

Data Collection → Data Cleaning → Exploratory Data Analysis → KPI Calculation → Visualization → Insights → Recommendations

6. Tools Used

Python, Pandas, NumPy, Matplotlib and Jupyter Notebook.

7. Analysis Performed

- Dataset shape and summary statistics
- Missing-value check
- Total sales, total profit and average order value
- Category-wise sales and profit
- Region-wise sales
- Monthly sales trend
- Product-wise sales

8. Visualizations

The analysis generates:

- Sales by Category
- Sales by Region
- Monthly Sales Trend

9. Business Insights

The code calculates the strongest categories, regions and products from the dataset. Monthly analysis can be used to identify periods of high and low demand. Sales and profit should be considered together because a category with high sales is not automatically the most profitable.

10. Recommendations

- Focus marketing on high-performing categories and products.
- Monitor low-performing regions and investigate customer demand.
- Use monthly trends for inventory planning.
- Evaluate discounts using profit as well as sales.
- Track product performance regularly to improve stock decisions.

11. Conclusion

This project demonstrates how raw e-commerce transaction data can be transformed into useful business information using Python. The analysis covers data inspection, KPI calculation, grouping, trend analysis and visualization, providing a practical foundation for data-driven decision making.

12. Future Scope

The project can be extended with customer segmentation, predictive sales forecasting, an interactive Power BI dashboard and machine-learning-based demand prediction.